Text Generation Using Hugging Face Inference API

1. AIM

To develop a Python program that uses a Hugging Face model through the Inference API to generate text from user-provided prompts and display the generated output.

2. PROCEDURE

1. Create an account on Hugging Face.
2. Generate an access token with permission to use Inference Providers.
3. Install the required Python library.
4. Import the InferenceClient class.
5. Connect to the Hugging Face Inference API using the token.
6. Accept a text prompt from the user.
7. Send the prompt to the model and display the generated response.

3. INSTALLATION

Open the terminal and execute:

pip install -U huggingface_hub

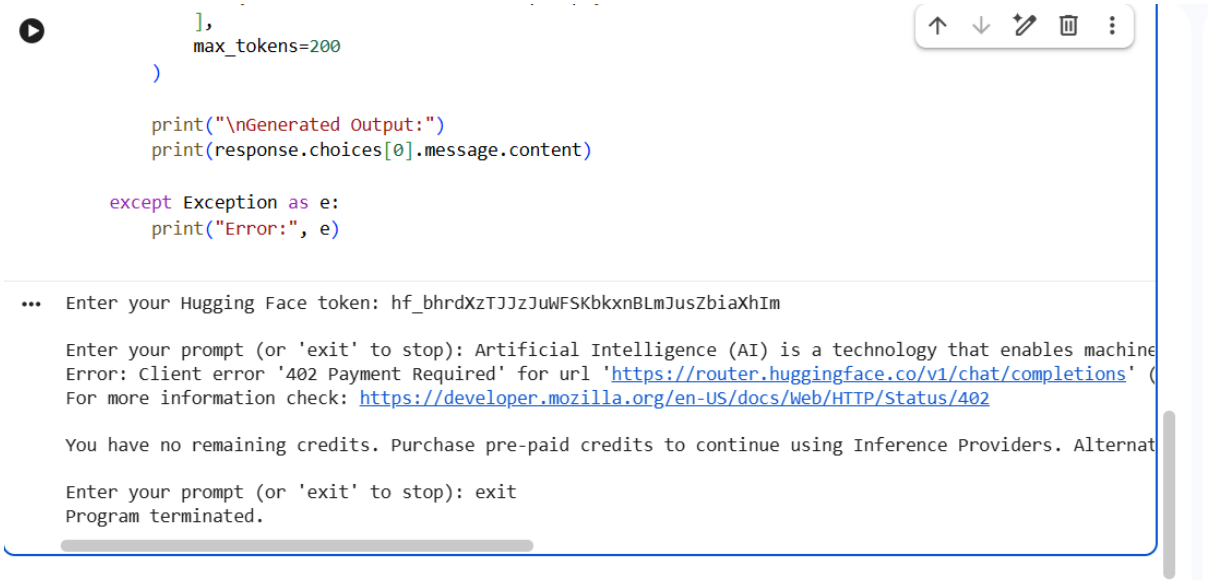
Create your access token at Hugging Face Settings – Tokens.

4. PYTHON PROGRAM

Save the following code as huggingface_text_generation.py.

from huggingface_hub import InferenceClienttoken = input("Enter your Hugging Face token: ")client = InferenceClient(api_key=token)while True: prompt = input("\nEnter your prompt (or 'exit' to stop): ") if prompt.lower() == "exit": print("Program terminated.") break try: response = client.chat.completions.create(model="openai/gpt-oss-120b", messages=[{"role": "user", "content": prompt}], max_tokens=200) print("\nGenerated Output:") print(response.choices[0].message.content) except Exception as e: print("Error:", e)

5. SAMPLE OUTPUT



```
            ],
            max_tokens=200
        )

        print("\nGenerated Output:")
        print(response.choices[0].message.content)

    except Exception as e:
        print("Error:", e)
```

```
Enter your Hugging Face token: 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

Enter your prompt (or 'exit' to stop): Artificial Intelligence (AI) is a technology that enables machine
Error: Client error '402 Payment Required' for url 'https://router.huggingface.co/v1/chat/completions' (
For more information check: https://developer.mozilla.org/en-US/docs/Web/HTTP/Status/402

You have no remaining credits. Purchase pre-paid credits to continue using Inference Providers. Alternat

Enter your prompt (or 'exit' to stop): exit
Program terminated.
```

6. RESULT

Thus, the Python program was developed successfully to accept user prompts, send them to a Hugging Face model through the Inference API, and display the generated text responses.